

Decision-dependent distributionally robust optimization for resilient and sustainable multimodal transportation under endogenous uncertainty

1 Impact of the chance-constraint risk level data

Table 1: DRTM results under different epsilon values

Order size	ϵ	Objective	β_{ij}^{mp}	α_{ij}^{msr}	U_l^s	D_l^s	GAP
10	0.005	580755	1	8	0	35	0.7137%
10	0.01	580836	1	8	0	35	0.7269%
10	0.015	579454	1	8	0	23	0.4920%
10	0.02	580908	1	6	1	23	0.9523%
10	0.025	580755	1	8	0	35	0.7137%
10	0.03	580967	1	8	0	0	0.7432%
10	0.035	580836	1	8	0	35	0.6859%
10	0.04	580908	1	6	1	23	0.9816%
10	0.045	579454	1	8	0	23	0.4920%
10	0.05	580755	1	8	0	35	0.7141%
10	0.055	580744	1	8	0	35	0.7131%
10	0.06	582364	1	8	15	0	0.9999%
10	0.065	580908	1	6	1	23	0.9797%
10	0.07	580908	1	6	1	23	0.9995%
10	0.075	580755	1	8	0	35	0.7137%
10	0.08	580908	1	6	1	23	0.9744%
10	0.085	580755	1	8	0	35	0.7137%
10	0.09	579454	1	8	0	23	0.4920%
10	0.095	580908	1	6	1	23	0.9959%
10	0.1	580755	1	8	0	35	0.7137%
30	0.005	1512313	3	10	6	0	0.9999%
30	0.01	1511804	3	10	6	0	0.9999%
30	0.015	1513656	3	9	0	4	0.9927%
30	0.02	1511321	3	10	6	0	0.9811%
30	0.025	1510980	3	9	6	0	0.9395%
30	0.03	1511793	3	10	6	0	0.9998%
30	0.035	1510733	3	10	6	0	0.9242%
30	0.04	1511613	3	10	6	0	1.0000%
30	0.045	1511779	3	10	6	0	0.9999%
30	0.05	1511783	3	10	6	0	0.9999%
30	0.055	1512183	3	10	6	0	0.9999%
30	0.06	1512174	3	10	6	0	0.9999%
30	0.065	1512600	3	10	6	0	0.9996%
30	0.07	1512179	3	10	6	0	0.9997%

Table 1: DRTM results under different epsilon values (continued)

Order size	ϵ	Objective	β_{ij}^{mp}	α_{ij}^{msr}	U_l^s	D_l^s	GAP
30	0.075	1511343	3	10	6	0	0.9857%
30	0.08	1511987	3	10	6	0	0.9964%
30	0.085	1509631	3	9	6	0	0.8509%
30	0.09	1509631	3	9	6	0	0.8504%
30	0.095	1511333	3	10	6	0	0.9816%
30	0.1	1513667	3	9	8	2	0.9999%

Table 2: D³RTM results under different epsilon values

Order size	ϵ	Objective	β_{ij}^{mp}	α_{ij}^{msr}	U_l^s	D_l^s	GAP
10	0.005	546373	1	7	1	23	0.9982%
10	0.01	546352	1	7	1	23	0.4804%
10	0.015	549404	3	3	0	0	0.9660%
10	0.02	547341	1	7	8	23	0.9805%
10	0.025	549404	3	3	0	0	0.9641%
10	0.03	549199	1	7	1	23	0.9348%
10	0.035	546349	1	7	1	23	0.9965%
10	0.04	547914	3	4	7	0	0.9998%
10	0.045	549404	3	3	0	0	0.9649%
10	0.05	546355	1	7	1	23	0.9961%
10	0.055	546357	1	7	1	23	0.4845%
10	0.06	546374	1	7	1	23	0.4906%
10	0.065	548333	1	8	15	23	0.8438%
10	0.07	546356	1	7	1	23	0.9965%
10	0.075	549404	3	3	0	0	0.9654%
10	0.08	549404	3	3	0	0	0.9506%
10	0.085	547371	1	7	8	23	0.9679%
10	0.09	547341	1	7	8	23	0.9978%
10	0.095	547341	1	7	8	23	0.6109%
10	0.1	547380	1	7	8	23	0.9934%
30	0.005	1422355	3	8	19	6	0.7916%
30	0.01	1425162	3	9	0	0	0.9173%
30	0.015	1420964	3	8	7	6	0.6835%
30	0.02	1424283	3	9	18	6	0.9191%
30	0.025	1422858	3	9	7	6	0.8339%
30	0.03	1423365	3	8	29	6	0.8826%
30	0.035	1425021	3	9	28	6	0.9827%
30	0.04	1424093	3	9	30	6	0.9169%
30	0.045	1422762	3	9	4	6	0.7968%
30	0.05	1420920	3	9	14	6	0.6999%
30	0.055	1424136	3	9	20	6	0.9171%
30	0.06	1423479	3	9	12	6	0.8800%
30	0.065	1421756	3	10	5	6	0.7627%
30	0.07	1425320	3	10	10	6	0.9874%
30	0.075	1424404	3	9	4	6	0.9422%
30	0.08	1421820	3	10	1	6	0.7584%
30	0.085	1420094	3	9	5	6	0.6401%
30	0.09	1420987	3	8	18	6	0.6945%

Table 2: D³RTM results under different epsilon values (continued)

Order size	ϵ	Objective	β_{ij}^{mp}	α_{ij}^{msr}	U_l^s	D_l^s	GAP
30	0.095	1425614	3	9	3	4	0.9450%
30	0.1	1425073	3	10	5	6	0.9947%